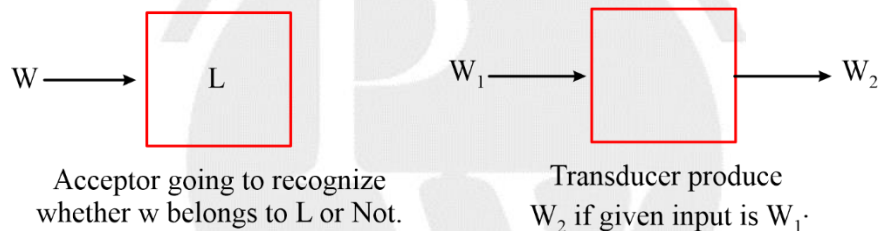


# 2

# FINITE AUTOMATA

## 2.1 Introduction

- Finite Automata describes or represents a regular language.
- Finite Automata is of two types:
  - Acceptor: Accepts or rejects given string.
  - Transducer: Produces output string for given input string.

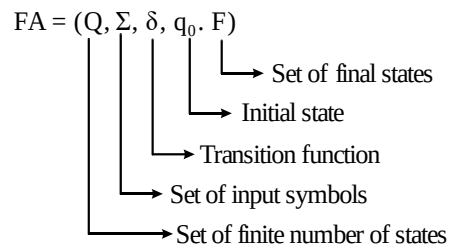


### Example:

1's complement of binary number: Input = 10100, Output = 01011

## 2.2 Finite Acceptor (Finite Automata)

- Finite Automata:

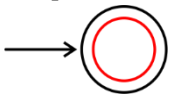


### Transition function ( $\delta$ )

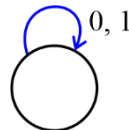
- (1) DFA :  $\delta: Q \times \Sigma, \rightarrow Q$
- (2) NFA:   
 Without  $\epsilon$ - moves  $\delta: Q \times \Sigma, \rightarrow 2^Q$    
 With  $\epsilon$ - moves  $\delta: Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$

## 2.3 Construction of DFA

1. If epsilon belongs to L, then initial state must be final in DFA.

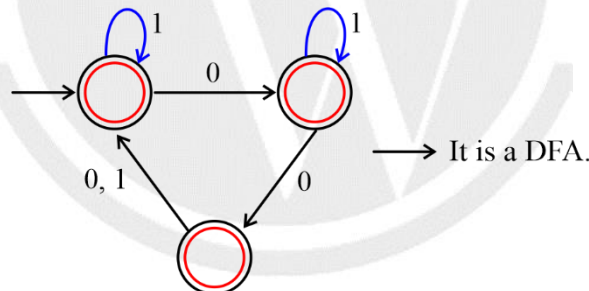


2. **Dead state:** It is non-final state but it never contain a path to final.

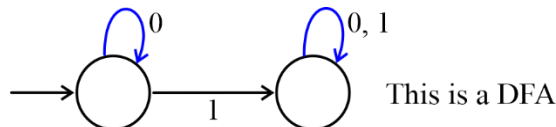


- Once we reach the dead state, there is no way to reach to the final state.
- If every state is final then every string accepted in the DFA.

3. If all states are finals in DFA then  $L(\text{DFA}) = \Sigma^*$



4. If every state is non final in a DFA then  $L(\text{DFA}) = \phi$  or  $L(\text{DFA}) = \{\}$ ,



$$\overline{L} = \Sigma^* - L$$

$$L \cup \overline{L} = \Sigma^*$$

$$L \cap \overline{L} = \phi$$

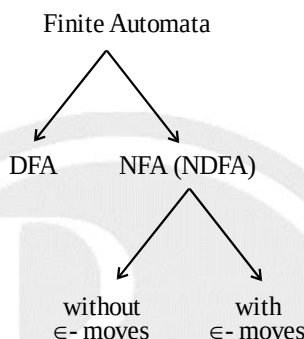
- If  $|w| = n$  or  $|w| \leq n$ , then number of states in minimum DFA =  $(n + 2)$  states
- If  $|w| \geq n$  then number of states in minimum DFA =  $n + 1$  states.
- Start condition, exactly, atleast length question requires Dead state but end condition, contain substring, atleast length questions do not require dead state.

- Over one symbol  $\Sigma = \{a\}$ . Length of string is equal to number of a's in string.  $|w| = n_a(w)$ .
- Let  $L = \{w | w \in \{a, b\}^*, n_a(w) \text{ is divisible by } m, n_b(w) \text{ is divisible by } n\}$ . Then Number of states in DFA =  $m \times n$ .
- Let  $L = \{w | w \text{ belongs to } \{a, b\}^*, n^{\text{th}} \text{ symbol of } w \text{ from begin is 'a'}\}$ . Then Number of states in DFA =  $(n + 2)$ .
- Number of equivalence classes for any regular set  $L$  = Number of states in minimal DFA that accepts  $L$ .

## 2.4 Non-Deterministic Finite Automata

### 2.4.1 Introduction

- Finite Automata can be designed in two ways:



- All these finite state machines are equivalent.  $DFA \cong NFA$
- We can convert one finite state machine to any other finite state machine.
- For every regular language, we can design infinite equivalent DFAs or infinite equivalent NFAs but minimum DFA is unique for given regular language.
- For every regular language, one or more minimum NFAs may exist.

**Note :** For every regular language:

- Unique minimum DFA exists.
- One or more minimum NFAs exists.

## 2.5 Comparison of NFA and DFA

|                                | $ w  = n$ | $ w  \leq n$ | $ w  \geq n$ |
|--------------------------------|-----------|--------------|--------------|
| <b>Number of states in NFA</b> | $n + 1$   | $n + 1$      | $n + 1$      |
| <b>Number of states in DFA</b> | $n + 2$   | $n + 2$      | $n + 1$      |

- If every string has  $n^{\text{th}}$  symbol from begin is 'a' over binary alphabet  $\{a, b\}$  then  $(n + 1)$  states in minimum NFA but  $(n+2)$  states in minimum DFA.

- If every string has  $n^{\text{th}}$  symbol from end is 'a' over binary alphabet  $\{a, b\}$  then  $2^n$  states in minimum DFA and  $(n + 1)$  states in minimum NFA.

### 2.5.1 COMPARISON OF DFA AND NFA (NFA vs DFA)

|                                      | NFA                                                            | DFA                                                    |
|--------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| (1) Transition Function ( $\delta$ ) | $Q \times \Sigma \rightarrow 2^Q$                              | $Q \times \Sigma \rightarrow Q$                        |
| (2) Number of paths for string       | For valid string: 1 path<br>For invalid string: $\geq 0$ paths | For valid string: 1 path<br>For invalid string: 1 path |

### 2.5.2 Number of states in DFA and NFA for Regular Languages

| Language                                                                         | NFA states | DFA states |
|----------------------------------------------------------------------------------|------------|------------|
| (1) $\{w \mid w \in \{a, b\}^*,  w  = n\}$                                       | $n + 1$    | $n + 2$    |
| (2) $\{w \mid w \in \{a, b\}^*,  w  \leq n\}$                                    | $n + 1$    | $n + 2$    |
| (3) $\{w \mid w \in \{a, b\}^*,  w  \geq n\}$                                    | $n + 1$    | $n + 2$    |
| (4) $\{w \mid w \in \{a, b\}^*, w \text{ starts with } a\}$                      | 2          | 3          |
| (5) $\{w \mid w \in \{a, b\}^*, n^{\text{th}} \text{ symbol from begin is } a\}$ | $n + 1$    | $n + 2$    |
| (6) $\{w \mid w \in \{a, b\}^*, n^{\text{th}} \text{ symbol from end is 'a'}\}$  | $n + 1$    | $2^n$      |

### 2.5.3 Finding number of states in DFA using NFA

NFA ( $n$  states)

↓ Subset Construction

DFA ( $2^n$  states exists)

↓ Partition algorithm

Minimum DFA ( $\leq 2^n$  states) atmost  $2^n$  states

- Every DFA is NFA, but NFA need not be DFA.

### 2.5.4 Relation between NFA with epsilon, NFA without epsilon, and DFA

